# Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Distance and midpoint

Find the distance between $A(1, 3)$ and $B(7, 11)$.

**2** [Foundational] • Target: Distance and midpoint

Find the midpoint of the segment joining $(-3, 5)$ and $(9, -1)$.

**3** [Foundational] • Target: Distance and midpoint

The midpoint of $A(2, -3)$ and $B(x, y)$ is $(5, 1)$. Find the coordinates of $B$.

**4** [Foundational] • Target: Gradient and equations of lines

Find the gradient of the line through $(2, -1)$ and $(6, 7)$.

**5** [Foundational] • Target: Gradient and equations of lines

Find the equation of the line through $(4, 3)$ with gradient $-2$, in the form $y = mx + c$.

**6** [Foundational] • Target: Gradient and equations of lines

Find the equation of the line through $(-1, 4)$ and $(3, -4)$, in the form $y = mx + c$.

**7** [Foundational] • Target: Parallel and perpendicular lines

Find the equation of the line parallel to $y = 3x - 1$ passing through $(5, 0)$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Parallel and perpendicular lines

Find the equation of the line perpendicular to $y = \frac{1}{3}x + 2$ passing through $(6, -1)$.

**9** [Intermediate] • Target: Parallel and perpendicular lines

Determine whether the lines $y = 4x - 3$ and $4y + x = 8$ are parallel, perpendicular, or neither.

**10** [Intermediate] • Target: The equation of a circle

Write the equation of a circle with center $(3, -2)$ and radius 6.

**11** [Intermediate] • Target: The equation of a circle

Find the center and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$ by completing the square.

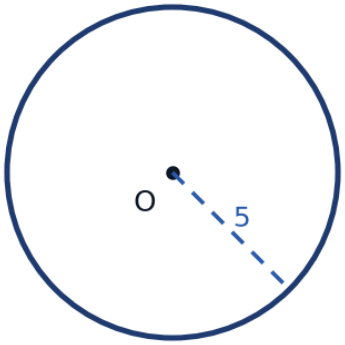
© ClipSAT — clipsat.org | All Rights Reserved

**12** [Intermediate] • Target: The equation of a circle

A circle has equation $(x + 1)^2 + (y - 4)^2 = 49$. State its center and radius.

**13** [Intermediate] • Target: Lines and circles together

A circle has center $(0, 0)$ and radius 5. Determine whether the line $y = x + 8$ intersects the circle, is tangent to it, or misses it entirely, by comparing the distance from the center to the line against the radius.




**14** [Intermediate] • Target: Lines and circles together

A circle has center $(2, 1)$ and radius 3. Determine whether the line $x + y = 10$ intersects the circle, is tangent to it, or misses it entirely.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: The perpendicular bisector • Cross-Lesson: 18.01 Binomial Distribution

Find the equation of the perpendicular bisector of the segment joining $A(1, 2)$ and $B(7, 8)$.

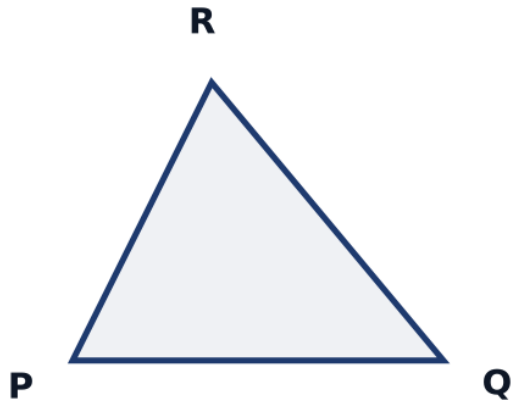
**16** [Challenging] • Target: The perpendicular bisector • Cross-Lesson: 19.01 Normal Distribution

Find the equation of the perpendicular bisector of the segment joining $A(-2, 3)$ and $B(4, -5)$.

© ClipSAT — clipsat.org | All Rights Reserved

**17** [Challenging] • Target: Area of a triangle from coordinates • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the area of the triangle with vertices $P(0, 0)$, $Q(8, 0)$, $R(3, 6)$.




**18** [Challenging] • Target: Area of a triangle from coordinates • Cross-Lesson: 3.01 Graphs & Transformations

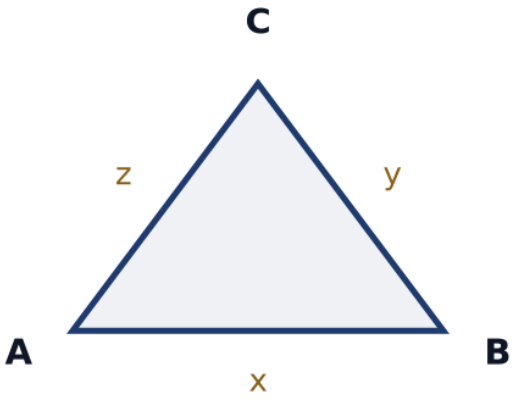
Find the area of the triangle with vertices $A(-3, 2)$, $B(5, 2)$, $C(1, 7)$.

**19** [Challenging] • Target: A well-designed word problem: locating a cell tower • Cross-Lesson: 5.01 Trigonometry

This question has two parts. Two houses are located at $A(0, 0)$ and $B(12, 0)$ (positions in kilometers). A cell tower must be built equidistant from both houses, and it must also lie on the line $y = x + 2$. (a) Find the equation of the perpendicular bisector of $AB$, and use it together with $y = x + 2$ to find the tower's coordinates. (b) Find the distance from the tower to each house, and confirm they are equal.

**20** [Challenging] • Target: A well-designed word problem: locating a cell tower • Cross-Lesson: 6.01 Differentiation & Integration

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.




## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

**21** [Foundational] • Target: Lines, Lengths & Midpoints

Find the equation of the line passing through $A(2, -3)$ and $B(6, 5)$ in the form $y = mx + c$.

**a.** $y = 2x - 7$

**b.** $y = 2x + 1$

**c.** $y = \frac{1}{2}x - 4$

**d.** $y = 2x - 3$

**22** [Foundational] • Target: Lines, Lengths & Midpoints

Find the length of the line segment joining $P(-1, 4)$ and $Q(5, -4)$.

**a.** 10

**b.** 8

**c.** $2\sqrt{7}$

**d.** 12

**23** [Foundational] • Target: Lines, Lengths & Midpoints

Find the midpoint of the segment connecting $C(-3, 8)$ and $D(7, -2)$.

**a.** $(2, 3)$

**b.** $(5, 3)$

**c.** $(2, 5)$

**d.** $(4, 6)$

**24** [Foundational] • Target: Perpendicular Bisectors & Gradients

Find the equation of the perpendicular bisector of the segment joining $A(1, 2)$ and $B(5, 10)$.

**a.** $x + 2y = 15$

**b.** $2x - y = 0$

**c.** $x - 2y = -9$

**d.** $2x + y = 12$

**25** [Foundational] • Target: Perpendicular Bisectors & Gradients

The line $L_1$ has equation $3x - 4y = 12$. Find the equation of the line $L_2$ perpendicular to $L_1$ passing through $(2, 1)$.

**a.** $4x + 3y = 11$

**b.** $3x + 4y = 10$

**c.** $4x - 3y = 5$

**d.** $3x - 4y = 2$

© ClipSAT — clipsat.org | All Rights Reserved

**26** [Foundational] • Target: Perpendicular Bisectors & Gradients

Three points are given: $P(1, 1)$, $Q(4, 5)$, and $R(k, 9)$. If $P, Q, R$ are collinear, find $k$.

**a.** 7

**b.** 6

**c.** 8

**d.** 5

**27** [Foundational] • Target: Circles — Equations, Tangents & Normals

Find the center and radius of the circle given by $x^2 + y^2 - 6x + 8y - 11 = 0$.

**a.** Center $(3, -4)$, radius 6

**b.** Center $(-3, 4)$, radius 6

**c.** Center $(3, -4)$, radius $\sqrt{11}$

**d.** Center $(-3, 4)$, radius 36

**28** [Foundational] • Target: Circles — Equations, Tangents & Normals

Find the equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

**a.** $3x + 4y = 25$

**b.** $4x - 3y = 0$

**c.** $3x - 4y = 25$

**d.** $4x + 3y = 24$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Circles — Equations, Tangents & Normals

A circle has diameter with endpoints $A(-2, 1)$ and $B(4, 9)$. Find the equation of the circle.

**a.** $(x - 1)^2 + (y - 5)^2 = 25$

**b.** $(x + 1)^2 + (y + 5)^2 = 25$

**c.** $(x - 1)^2 + (y - 5)^2 = 100$

**d.** $(x - 3)^2 + (y - 4)^2 = 25$

© ClipSAT — clipsat.org | All Rights Reserved

**30** **[Intermediate]** • **Target: Intersections & Exam-Standard Geometry**

Determine the relationship between the line $y = x - 5$ and the circle $x^2 + y^2 = 9$.

**a.** The line does not intersect the circle

**b.** The line is tangent to the circle

**c.** The line intersects the circle in two distinct points

**d.** The line passes through the center of the circle

**31** **[Intermediate]** • **Target: Intersections & Exam-Standard Geometry**

Find the area of the triangle with vertices $A(0, 0)$, $B(6, 2)$, and $C(2, 8)$.

**a.** 22

**b.** 20

**c.** 24

**d.** 18

**32** **[Intermediate]** • **Target: Intersections & Exam-Standard Geometry**

Find the length of the chord cut off from the line $y = 2$ by the circle $x^2 + y^2 = 13$.

**a.** 6

**b.** 3

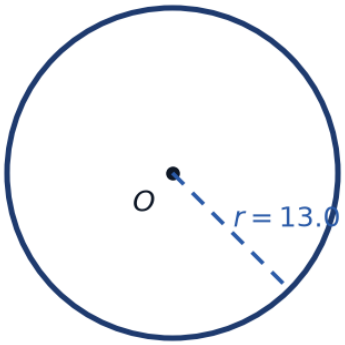
**c.** $2\sqrt{13}$

**d.** 8

© ClipSAT — clipsat.org | All Rights Reserved

**33** [Intermediate] • Target: Distance and midpoint

A circle has equation $(x-7)^2+(y+2)^2=169$. State its centre and radius.




**a.** centre $(7,-2)$, radius $13$

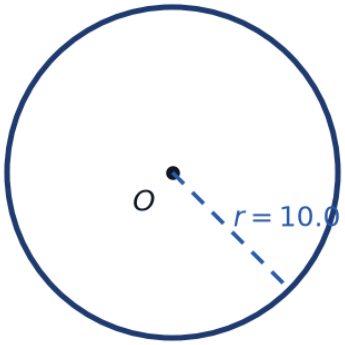
**b.** centre $(-2,7)$, radius $13$

**c.** centre $(-7,2)$, radius $13$

**d.** centre $(7,-2)$, radius $169$

**34** [Intermediate] • Target: Gradient and equations of lines

A circle has equation $(x+5)^2+(y-3)^2=100$. State its centre and radius.




**a.** centre $(-5,3)$, radius $10$

**b.** centre $(-5,3)$, radius $100$

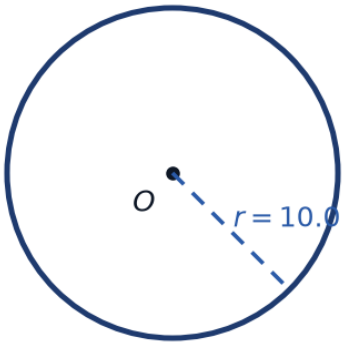
**c.** centre $(3,-5)$, radius $10$

**d.** centre $(5,-3)$, radius $10$

© ClipSAT — clipsat.org | All Rights Reserved

**35** [Intermediate] • Target: Parallel and perpendicular lines

A circle has equation $(x-3)^2+(y+3)^2=100$. State its centre and radius.




**a.** centre $(3, -3)$, radius $10$

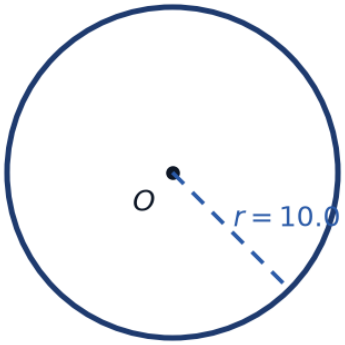
**b.** centre $(4, -3)$, radius $10$

**c.** centre $(3, -3)$, radius $100$

**d.** centre $(-3, 3)$, radius $10$

**36** [Intermediate] • Target: The equation of a circle

A circle has equation $(x-3)^2+(y+5)^2=100$. State its centre and radius.




**a.** centre $(-3, 5)$, radius $10$

**b.** centre $(3, -5)$, radius $10$

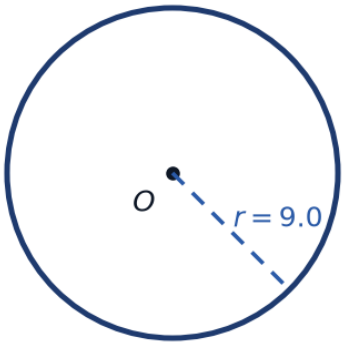
**c.** centre $(-5, 3)$, radius $10$

**d.** centre $(3, -5)$, radius $100$

© ClipSAT — clipsat.org | All Rights Reserved

**37** [Intermediate] • Target: Lines and circles together

A circle has equation $(x-5)^2+(y+5)^2=81$. State its centre and radius.




**a.** centre $(5,-5)$, radius $81$

**b.** centre $(5,-5)$, radius $9$

**c.** centre $(-5,5)$, radius $9$

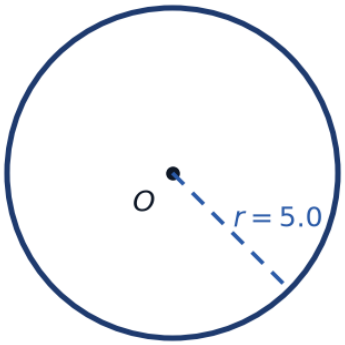
**d.** centre $(6,-5)$, radius $9$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

**38** [Challenging] • Target: The perpendicular bisector • Cross-Lesson: 3.01 Graphs & Transformations

A circle has equation $(x + 1)^2 + (y - 7)^2 = 25$. State its centre and radius.




**a.** centre $(7, -1)$, radius $5$

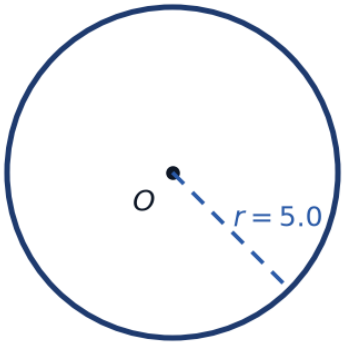
**b.** centre $(-1, 7)$, radius $25$

**c.** centre $(1, -7)$, radius $5$

**d.** centre $(-1, 7)$, radius $5$

**39** [Challenging] • Target: Area of a triangle from coordinates • Cross-Lesson: 5.01 Trigonometry

A circle has equation $(x - 7)^2 + (y - 5)^2 = 25$. State its centre and radius.




**a.** centre $(5, 7)$, radius $5$

**b.** centre $(7, 5)$, radius $5$

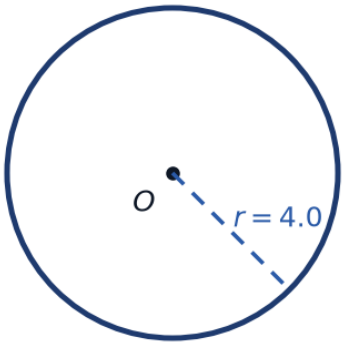
**c.** centre $(-7, -5)$, radius $5$

**d.** centre $(7, 5)$, radius $25$

© ClipSAT — clipsat.org | All Rights Reserved

**40** [Challenging] • Target: A well-designed word problem: locating a cell tower • Cross-Lesson: 6.01 Differentiation & Integration

A circle has equation $(x-7)^2+(y-8)^2=16$. State its centre and radius.




**a.** centre $(7, 8)$, radius $16$

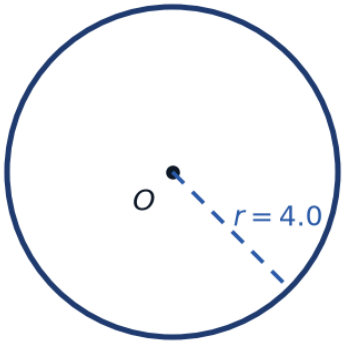
**b.** centre $(-7, -8)$, radius $4$

**c.** centre $(7, 8)$, radius $4$

**d.** centre $(8, 7)$, radius $4$

**41** [Challenging] • Target: Lines, Lengths & Midpoints • Cross-Lesson: 7.01 Statistics & Probability

A circle has equation $(x-6)^2+(y-5)^2=16$. State its centre and radius.




**a.** centre $(6, 5)$, radius $16$

**b.** centre $(-6, -5)$, radius $4$

**c.** centre $(5, 6)$, radius $4$

**d.** centre $(6, 5)$, radius $4$

© ClipSAT — clipsat.org | All Rights Reserved

**42** [Challenging] • Target: Perpendicular Bisectors & Gradients • Cross-Lesson: 8.01 Functions: Composite & Inverse

Find the equation of the circle with centre $(-7, -2)$ and radius $12$.

**a.** $(x+7)^2+(y+2)^2=144$

**b.** $(x-7)^2+(y-2)^2=144$

**c.** $(x+7)^2+(y+2)^2=12$

**d.** $(x+7)^2+(y+2)^2=148$

**43** [Challenging] • Target: Circles — Equations, Tangents & Normals • Cross-Lesson: 9.01 Circular Measure

Find the equation of the circle with centre $(-5, 2)$ and radius $11$.

**a.** $(x+5)^2+(y-2)^2=125$

**b.** $(x+5)^2+(y-2)^2=11$

**c.** $(x-5)^2+(y+2)^2=121$

**d.** $(x+5)^2+(y-2)^2=121$

**44** [Challenging] • Target: Intersections & Exam-Standard Geometry • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Find the equation of the circle with centre $(-3, -3)$ and radius $4$.

**a.** $(x+3)^2+(y+3)^2=20$

**b.** $(x+3)^2+(y+3)^2=4$

**c.** $(x+3)^2+(y+3)^2=16$

**d.** $(x-3)^2+(y-3)^2=16$

**45** [Challenging] • Target: Distance and midpoint • Cross-Lesson: 10.01 Series & Binomial Theorem

Find the equation of the circle with centre $(-2, -6)$ and radius $6$.

**a.** $(x+2)^2+(y+6)^2=36$

**b.** $(x+2)^2+(y+6)^2=40$

**c.** $(x+2)^2+(y+6)^2=6$

**d.** $(x-2)^2+(y-6)^2=36$

© ClipSAT — clipsat.org | All Rights Reserved